Characterising the complexity of time series network graphs: a simplicial approach

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Outline

- Time series and networks
- The visibility algorithm
- Simplicial complexes
- Topological characterisers
- Application to the logistic map
- Application to networks
- Outlook

Time series Analysis/ Networks Analysis

- The analysis of time series is well established
- Useful for analysing the dynamical behaviour of a system
- A variety of well developed tools and precise metrics exist: e.g. Fourier transforms, dimensions and entropies, Lyapunov exponents.
- Networks are a well established paradigms in complex systems
- A number of well established metrics exist for network analysis. These include path lengths, clustering co-efficients, degree distributions etc

Time series networks

- The network representation of time series yields new insights into the dynamical behaviour of system

- We use the methods of algebraic topology to define a new set of characterisers for the time series networks, which go beyond the usual network characterisers.

- We demonstrate the utility of the characterisers in the context of the time series arising from the logistic map and traffic on a network.

- First, the visibility algorithm is used to map the time series arising from the logistic map to a network.

The visibility algorithm (Lacasa , Lucque, 2008)

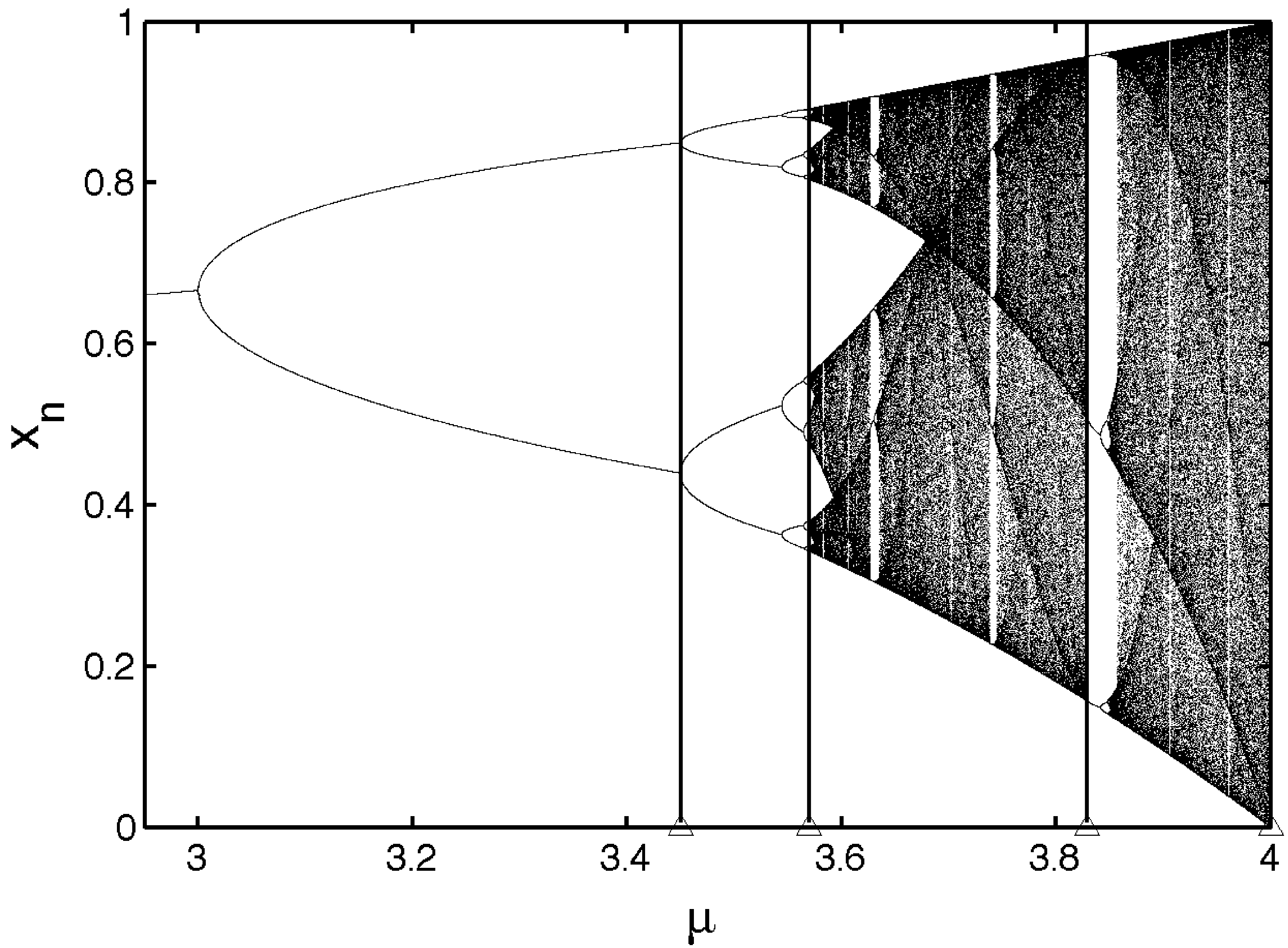
- Let (y_i, t_i) and (y_j, t_j) be two points in the time series $(y_i, t_i)\ i = 1 \ldots N$

- For these two points to be visible to each other, all other intermediate points should satisfy

$$y_j > y_r + \frac{y_j - y_i}{t_j - t_i}(t_j - t_r)$$

- If two points are visible to each other they are connected by a line. Thus a network equivalent to a time series is built up.

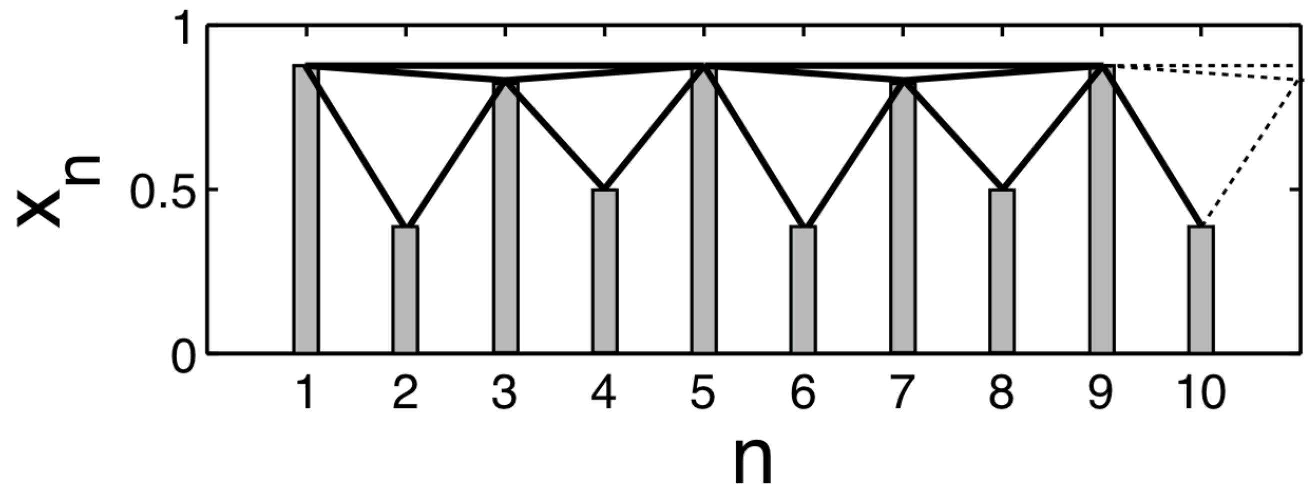
The logistic map: Bifurcation diagram

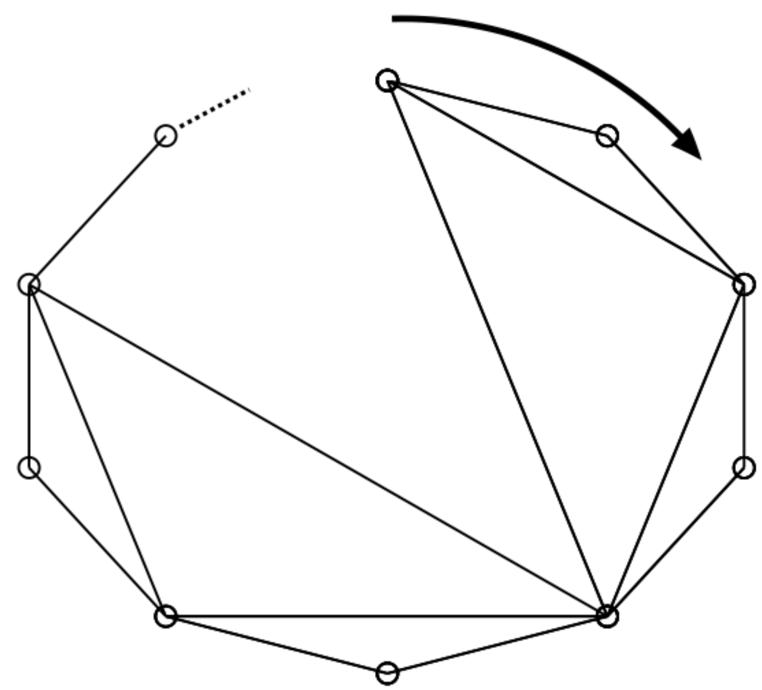


The TS Network

$\mu = 3.5$

- Periodic time series at $\mu = 3.5$



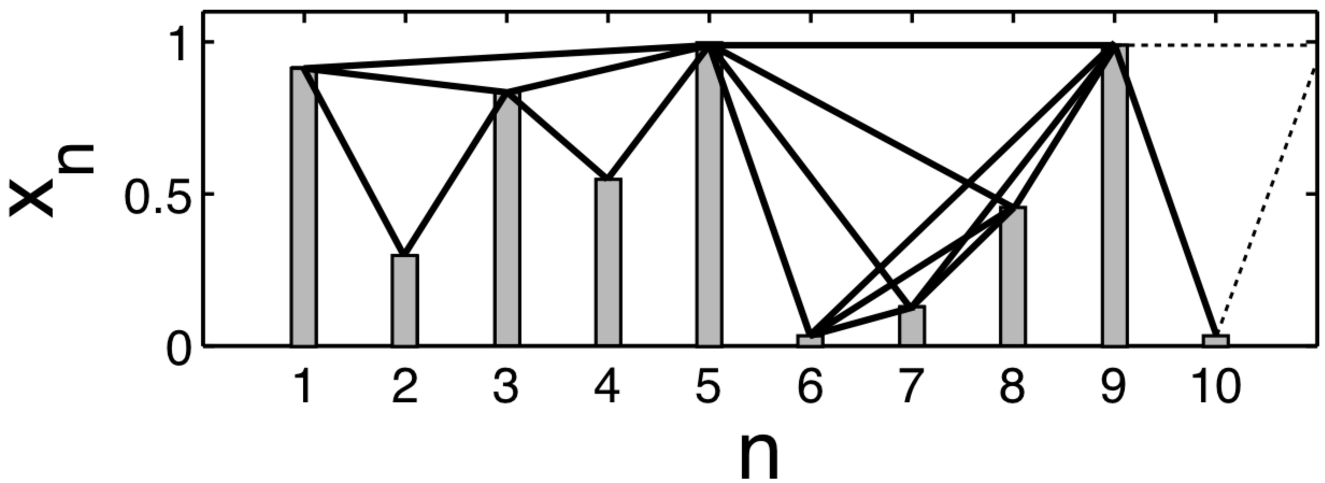


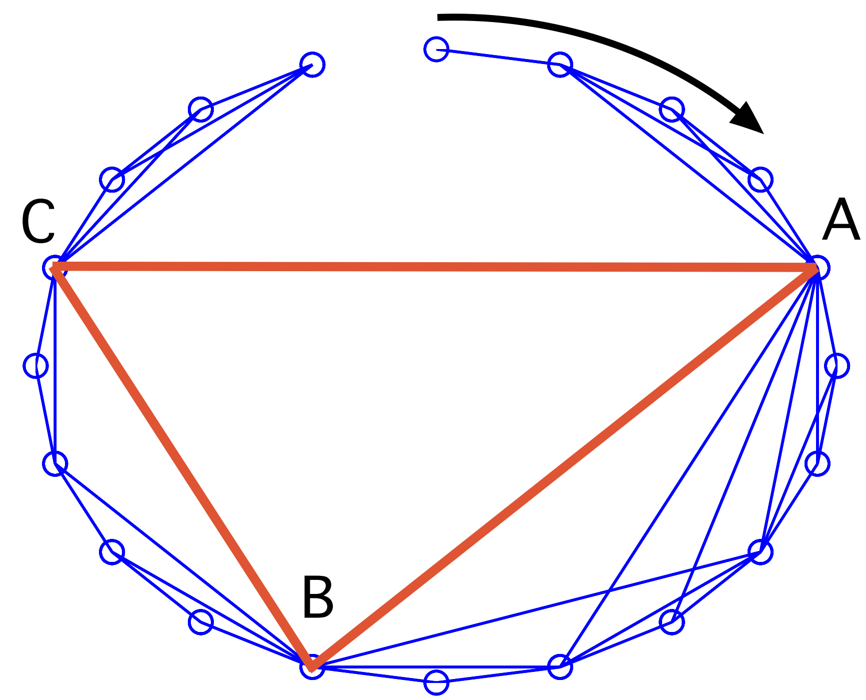
TS Network

$\mu = 4.0$

- TS network

$\mu = 4.0$





Simplicial analysis of TS networks

- A simplex with q+1 nodes or vertices is a q-dimensional simplex.
- 0-simplex, isolated point.
- 1-simplex, two vertices connected by a line segment
- 2-simplex: Triangle of three connected nodes
- 3-simplex: Tetrahedron with 4 connected nodes
- If two simplices have q+1 nodes in common they share a q-face.

Simplicial analysis of cliques

- Consider a simplicial complex where the simplices are cliques.
- A clique is a maximal complete sub-graph.
- We use the Bron-Kerbosch algorithm to find cliques using the connectivity matrix of the network.
- The cliques are analysed using characterisers from algebraic topology.

Topological characterisers (global)

- First structure vector Q: The q-th component of the vector $\mathbf{Q} = \{Q_0, Q_1, \ldots Q_{max}\}$ is the number of q-connected components at the q-th level.
- The second quantity, the vector $\bar{\mathbf{f}}$ has the number of q-dimensional simplices as its q-th component.
- The third quantity, the second structure vector $\mathbf{N_s} = \{n_0, n_1, \ldots n_{qmax}\}$ has the number of simplices of dimension q and higher as its q-th component
- The third structure vector, $\hat{\mathbf{Q}}$ has as its q-th component

$$\hat{Q}_q = (1 - \frac{Q_q}{n_q})$$

Topological characterisers (global)

- Reference:
- J. Jonsson, Simplicial Complexes of Graphs, Lecture Notes in Mathematics, Springer-Verlag, Berlin, Heidelberg (2008).

Topological characterisers (local)

- A fifth quantity $dim\ Q^i$ defines the topological dimension of a node i of the simplicial complex, given by

$$dim\ Q^i = \sum_{q=0}^{q_{max}} Q^i_q$$

- The topological entropy is defined by

$$S_Q(q) = -\frac{\sum_i p^i_q log p^i_q}{log N_q}$$

- Here, $p^i_q = \frac{Q^i_q}{\sum_i Q^i_q}$ $N_q = \sum_i (1 - \delta_{Q^i_q,0})$

Topological characterisers (local)

- References:

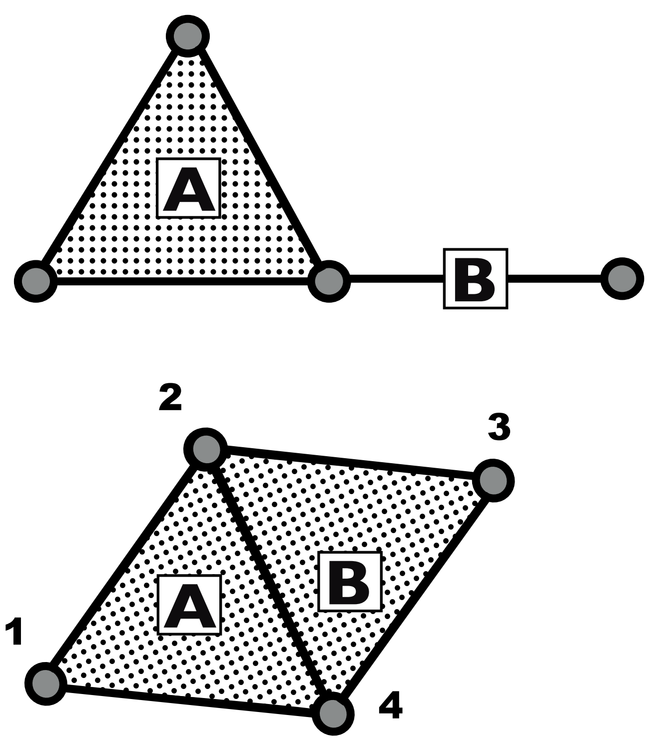
- M. Andjelkovic, B. Tadic, S. Maletic, M. Rajkovic, Hierachical sequencing of on-line social graphs, Physica A, vol. 436, 582-595 (2014).

- M. Andjelkovic, N. Gupte, B. Tadic, Hidden geometry of traffic jamming, Phys. Rev. E, 91, 052817 (2015).

Example

i)

ii)



- Two simplicial complexes.

Example

- Fig ii has 2 simplices A={1,2,4} and B={2,3,4}
- The incidence matrix is given by

$$\mathbf{\Lambda} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

- Both simplices have 3 nodes, therefore they have dimension 2.
- They have 2 nodes in common, therefore they share a 1-face.
- They are 1-connected, and therefore also 0-connected. The simplicial complex at the q=0, q=1, levels form a single entity. Hence Q_0 =1= Q_1

Example

- The simplices A and B are not connected at the q=2 level, since they do not have 3 points in common. Therefore the complex has two separate entitities at the q=2 level, and Q2=2
- Thus $\mathbf{Q}$ =(1,1,2)
- q-th component of $\bar{\mathbf{f}}$ is the number of simplices of dimension q
- Here, there are no simplices of 0 or 1 level, and there are 2 simplices of level 2
- therefore $\bar{\mathbf{f}}$ =(0,0,2)

Example: Structure vector N_s, Qhat

- The q-th component of the second structure vector, N_s is the number of simplices of dimension q and higher.
- Hence N_s =(2,2,2)
- The q-th component of the third structure vector $\hat{Q}$ is $(1 - \frac{Q_q}{n_q})$
- Hence $\hat{Q}$ =(1/2,1/2,0)
- qth component of Q^i vector: no of q-dimensional simplices in which node i participates.
- Hence Q^1=(0,0,1), Q^3=(0,0,1), Q^2=(0,0,2) $= Q^4$;

Example : Local quantities

- Hence dim Q^1=1; dim Q^2=2dimQ^3 =1, dim Q^4=2. Hence, Max dim Qi=2, for this complex

- We defined $p_q^i = Q_q^i / \sum_i Q_q^i$ $S_Q(q) = -\frac{\sum_i p_q^i \log p_q^i}{\log N_q}$ $N_q = \sum_i \left(1 - \delta_{Q_q^i,0}\right)$

- Only q=2 has non-zero probabilities, $p_2^2 = p_2^4 = 1/3$

$$p_2^1 = p_2^3 = 1/6$$

$$S_Q(2) = 0.9591$$

Entropy vector

$$S_Q(0) = 0 \qquad\qquad S_Q(1) = 0$$

- Therefore, $\mathbf{S} = (0, 0, 0.9591)$

Topological characterisers for the logistic map

	$\mu = 3.45$			$\mu = 3.56995$			$\mu = 3.82843$			$\mu = 4.0$		
q-level	$\mathbf{Q}$	$\mathbf{N_s}$	$\widehat{\mathbf{Q}}$	$\mathbf{Q}$	$\mathbf{N_s}$	$\widehat{\mathbf{Q}}$	$\mathbf{Q}$	$\mathbf{N_s}$	$\widehat{\mathbf{Q}}$	$\mathbf{Q}$	$\mathbf{N_s}$	$\widehat{\mathbf{Q}}$
0	1	1499	0.999	1	1988	0.999	1	1333	0.999	1	1984	0.999
1	501	1499	0.665	12	1988	0.993	667	1333	0.499	18	1984	0.990
2	1498	1498	0	1987	1987	0	1332	1332	0	1981	1981	0

	$\mu = 3.45$		$\mu = 3.56995$		$\mu = 3.82843$		$\mu = 4.0$	
q-level	$\mathbf{S}$	$\tilde{\mathbf{f}}$	$\mathbf{S}$	$\tilde{\mathbf{f}}$	$\mathbf{S}$	$\tilde{\mathbf{f}}$	$\mathbf{S}$	$\tilde{\mathbf{f}}$
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	0.95915	3
2	0.97745	1498	0.95455	1987	0.98852	1332	0.95506	1981

Max dimension of node

- Parameter versus maximum dimension

μ	$\dim Q^i$
3.45000	4
3.56995	19
3.82843	3
4.00000	25

Observations

- The topological structure of the simplicial complex is most easily identified by the vector $\bar{\mathbf{f}}$ which counts the simplices at each level. Here almost all simplices are at the level 2, and the number of simplices for the chaotic regimes (μ = 3.569995, 4.0) (1987, 1981) is higher than those at the periodic regimes (μ = 3.45, 3.82843) (1498, 1332).

- The vector $\mathbf{Q}$ counts the number of connected components at each topology level. For q=0, Q_0 =1, i.e. no isolated nodes. For q=1, chaotic states have fewer disconnected components (12,18) than periodic states (501,667). Therefore chaotic states are more connected. At q=2, also it is clear that chaotic states yield higher dimensional simplices

Observations

- max dim Q^i is the number of simplices in which the most connected node participates. For the periodic regime, this is 4 and 3. For the chaotic regimes, this is 19 and 25. Entropies are also lower for the chaotic states.

- All characterisers consistently indicate that networks representing chaotic states are more connected than networks representing periodic states.

- Local quantities pick up the differences more sharply

Application to a traffic network

- Simplicial characterisers applied to a model of traffic on a network. (Work done in collaboration with B. Tadic and M. Andjelkovic)

- The network is the Webgraph, which is characterized by two hubs and scale free degree distributions which are different for the outgoing and incoming links. The network has a finite clustering co-efficient and disassortative mixing.

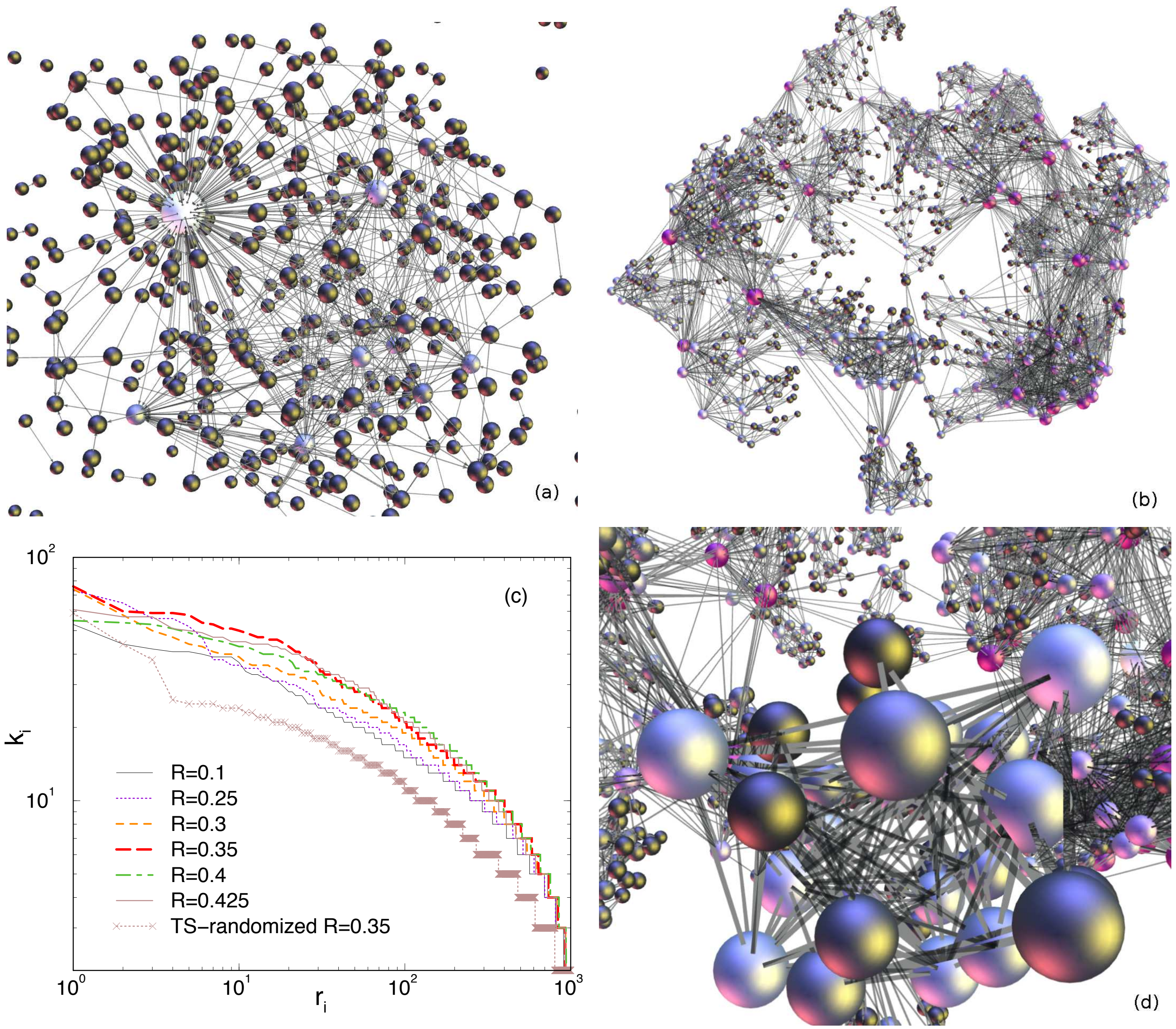
- We generate the traffic time series by performing simulations of the transport of information packets in dynamically distinct regimes: Free flow, jamming and congested.

Transmission of packets

- Guided walks from origin to destination using next nearest neighbour search rule. Packets form ordered queues at nodes. LIFO rule of transmission.
- Network size 1000 nodes, maximum queue length 1000.
- Traffic density controlled by a control parameter R, the probability of a new packet being generated at a node per time step.
- No. of packets on net $N_p(t)$
- No of active queues $n_g(t)$
- Two time series generated in each distinct traffic regime.

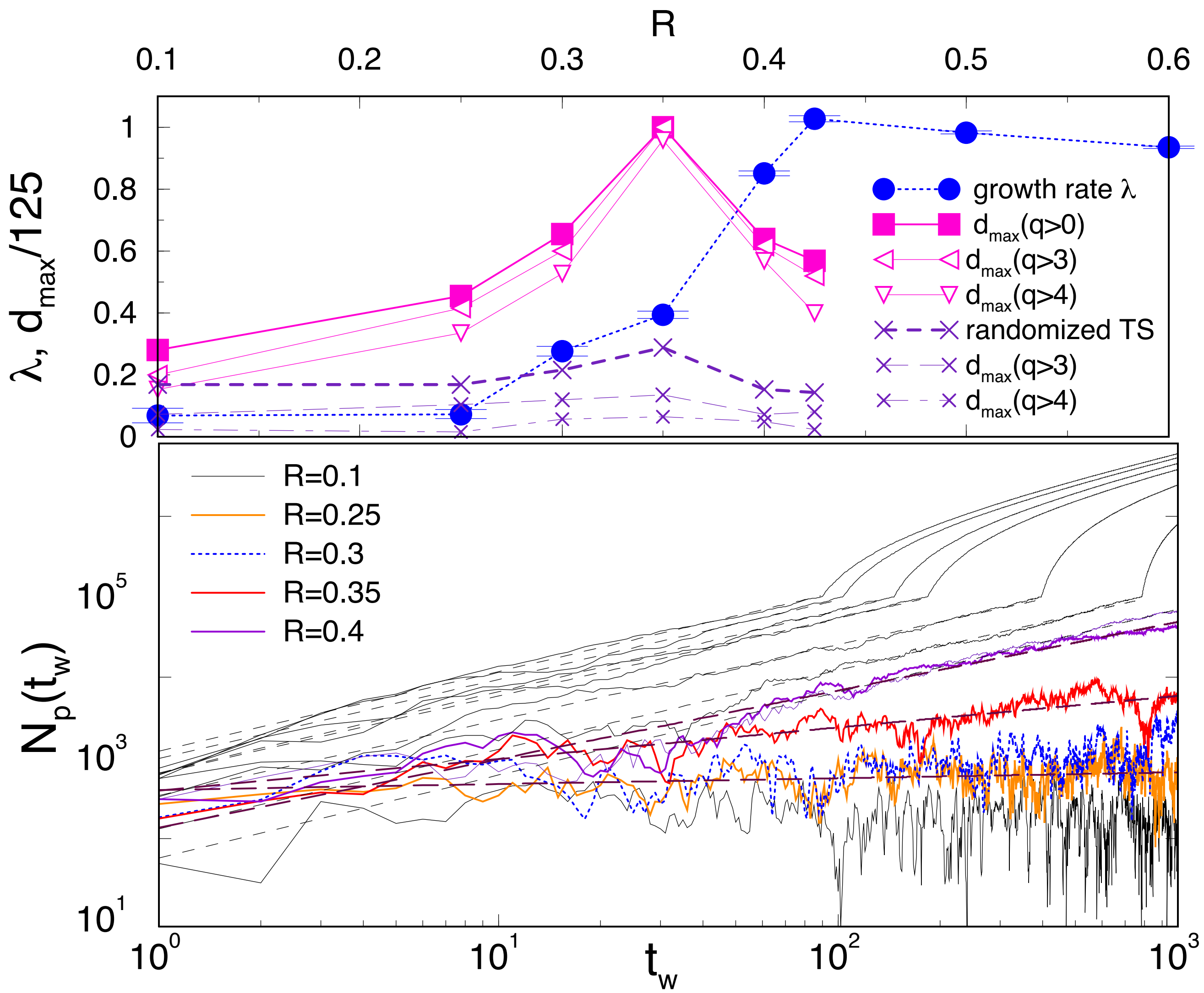
All the traffic networks

- (a) Substrate (b)TS net (c) degree (d) closeup of TS net



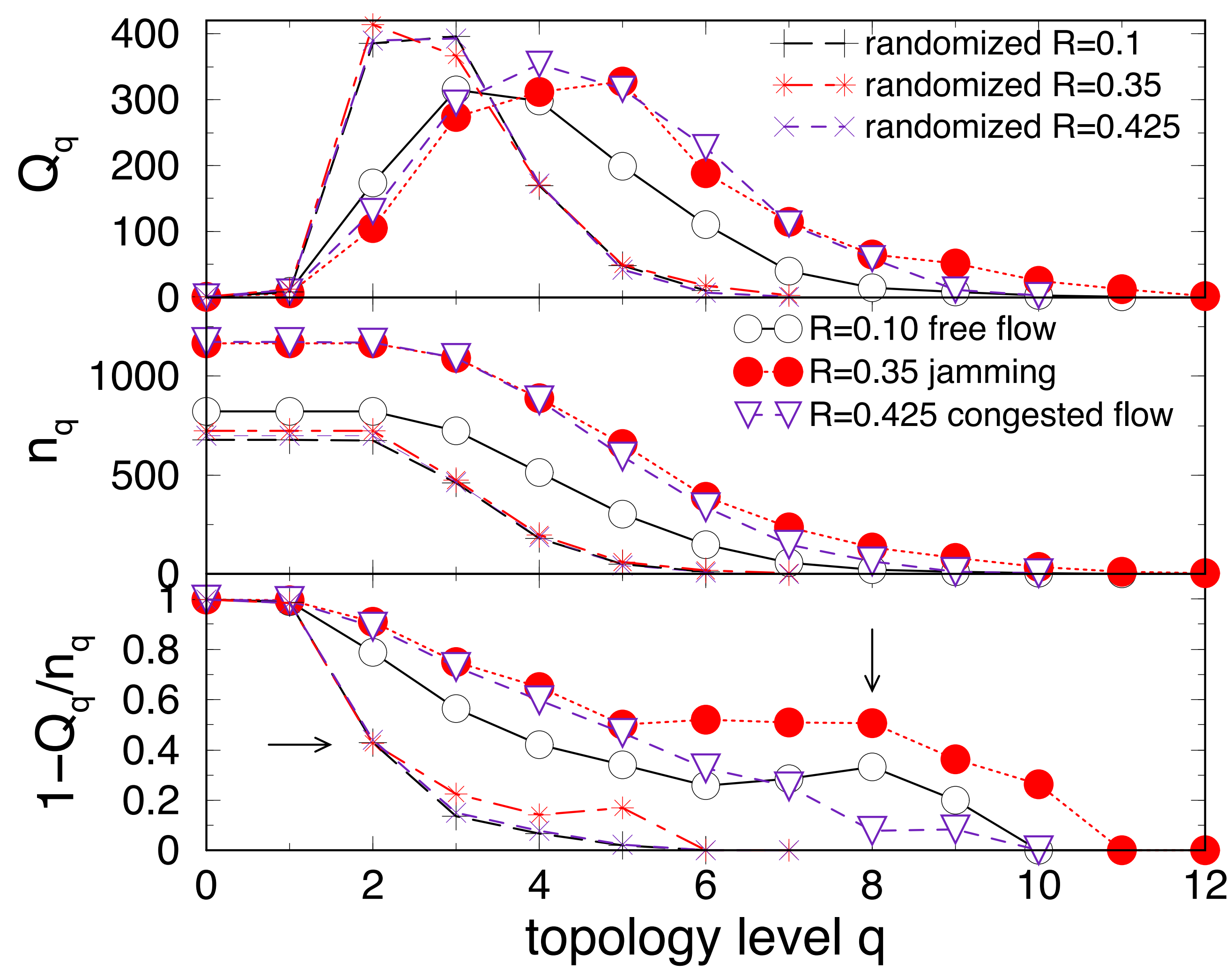
Time series and characterisers

- Max dimension and characterisers



Traffic quantities

- First, second and third structure vectors.



Entropy for traffic

- f(q) and S(q)

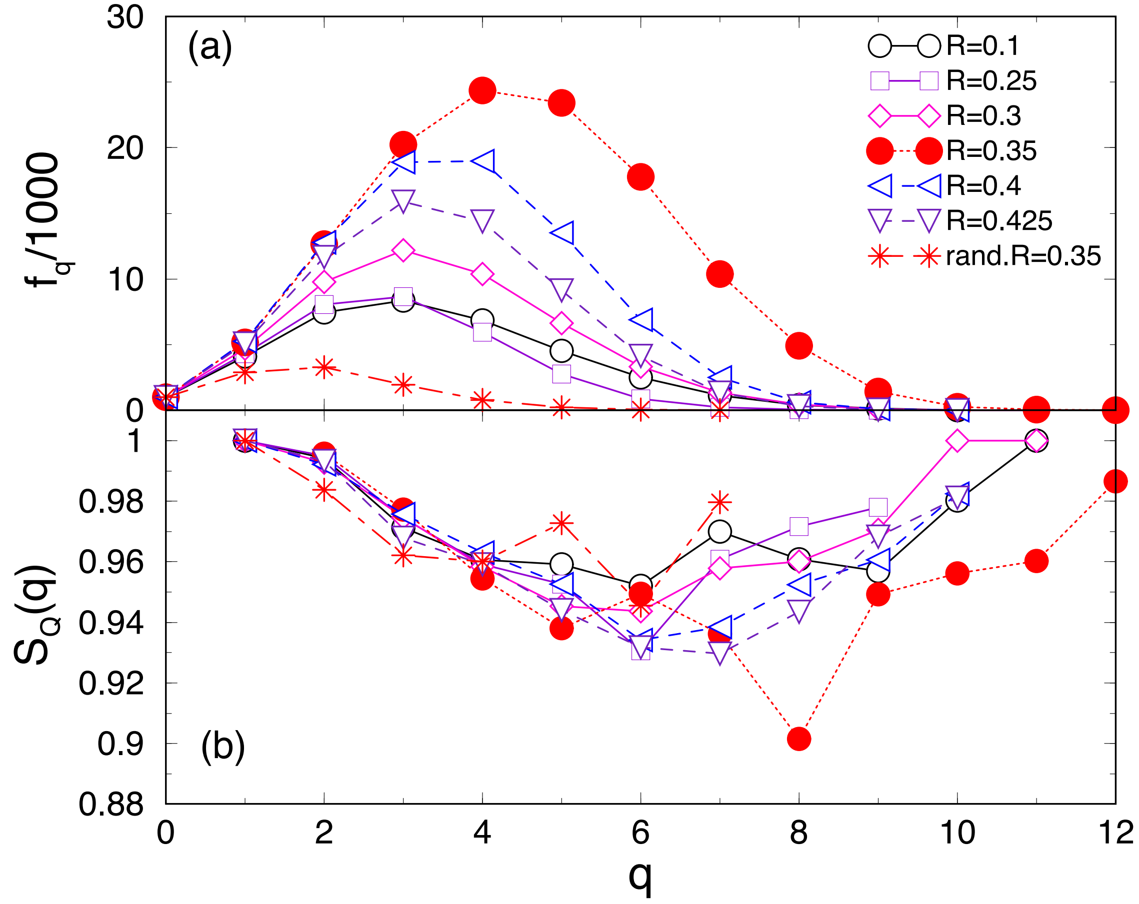


Table of characterisers

- 2 R values in each regime: free flow, jamming, congested

	R=0.1			**R=0.25**			**R=0.3**			**R=0.35**			**R=0.4**			**R=0.425**			randomized **R=0.35**		
q	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$	Q	N_s	$\hat{Q}$
0	1	823	0.99	1	906	0.99	1	979	0.99	1	1168	0.99	1	1148	0.99	1	1174	0.99	1	724	0.99
1	9	823	0.99	8	906	0.99	4	979	0.99	6	1168	0.99	9	1148	0.99	8	1174	0.99	12	724	0.98
2	174	822	0.79	136	905	0.85	117	978	0.88	105	1168	0.91	103	1147	0.91	130	1172	0.89	414	723	0.43
3	315	723	0.56	296	823	0.64	275	908	0.7	274	1093	0.75	260	1078	0.76	295	1094	0.73	367	473	0.22
4	298	515	0.42	340	622	0.45	310	715	0.57	312	891	0.65	298	888	0.66	354	877	0.6	170	198	0.14
5	199	302	0.34	235	362	0.35	307	485	0.37	328	657	0.50	292	654	0.55	316	595	0.47	49	59	0.7
6	110	148	0.26	152	177	0.14	171	238	0.28	188	392	0.52	259	422	0.39	227	337	0.33	18	18	0
7	40	56	0.29	43	49	0.12	74	94	0.21	115	234	0.51	149	208	0.28	111	149	0.26	3	3	0
8	14	21	0.33	7	9	0.22	23	32	0.28	65	132	0.51	68	87	0.22	59	64	0.08	-	-	-
9	8	10	0.2	3	3	0	11	12	0.08	51	80	0.36	31	32	0.03	11	12	0.08	-	-	-
10	3	3	0	-	-	-	3	3	0	25	34	0.26	6	6	0	3	3	0	-	-	-
11	1	1	0	-	-	-	1	1	0	12	12	0	-	-	-	-	-	-	-	-	-
12	-	-	-	-	-	-	-	-	-	2	2	0	-	-	-	-	-	-	-	-	-

Topological characterisers and the edge of jamming

- The order parameter $\lambda = \frac{log \;\; N_p(t_w)}{log \;\; (t_w)}$ shows a steep rise at R=0.35.

- Thus R=0.35 is the edge of jamming.

- The topological characterisers calculate using the TS net at R=0.35, show maximal complexity.

- The vertex maximum topological dimension also peaks at R=0.35.

- The lowest entropy is discovered for the most complex structure, again at R-0.35, at q=8.

Topological characterisers at the edge of jamming

- The highest peak in the f_q values is also seen at R=0.35.
- We note that these features are lost or reduced for the randomised series.
- Thus the edge of jamming is accurately picked up by the new chracterisers, especially the new local ones proposed by us.

Conclusions

- In the case of the logistic map, the utility of algebraic characterisers is demonstrated in a simple context where the dynamical behaviour is well understood.
- In the case of the traffic network, the utility in picking up the edge of jamming is demonstrated.
- The algebraic characterisers thus look like promising candidates for revealing the hidden geometry of networks with nontrivial correlations between dynamical states.
- May be useful for the analysis of crisis, intermittency, unstable dimension variability and others.